Bioenergetics

Store light energy

1 Photosynthesis

Carbon dioxide + water → glucose + oxygen. Light transfers energy to chloroplasts; chlorophyll absorbs it.

Endothermic: energy is taken in. Plants respire all the time; photosynthesis needs light.

2 Use the glucose

Respiration; starch and fat/oil storage; cellulose cell walls.

Glucose plus nitrate-derived nitrogen → amino acids → proteins.

Control & measure

3 Limiting factors

Relative light intensity: 4 → 1

Light, CO_2, temperature and chlorophyll can limit rate. A plateau means extra light alone will not help.

Intensity ∝ 1/distance²; doubling distance quarters intensity.

Greenhouses: balance extra crop income against heat/light/CO_2 costs.

4 Pondweed practical

Change lamp distance; measure oxygen volume/time. Control temperature, CO_2 supply, plant and background light.

Acclimatise; repeat; calculate mean. Bubbles vary in size, so volume is usually better.

Rate = volume/time; plot against light intensity or 1/distance^2.

Release & use energy

5 Respiration pathways

Aerobic: glucose + O_2 → CO_2 + water; more energy transferred.

Anaerobic: muscles → lactic acid; yeast/plants → ethanol + CO_2. Less energy per glucose.

Respiration is exothermic and continuous; fermentation makes bread rise and produces alcohol.

6 Exercise & metabolism

Exercise raises heart rate, breathing rate/volume. Oxygen shortage → more anaerobic respiration and fatigue.

Oxygen debt: extra oxygen needed afterwards. Blood takes lactic acid to liver → glucose.

Metabolism = all cell/body reactions; builds carbohydrates, lipids and proteins; excess nitrogen → urea.